

## 고완풀 단원별 급수 2단원 해설

$$\begin{aligned}
 \text{[1] A. } & \frac{\sqrt{3}}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left(x - \frac{\pi}{3}\right)^{2n} + \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \left(x - \frac{\pi}{3}\right)^{2n+1} \\
 &= \frac{\sqrt{3}}{2} \cos\left(x - \frac{\pi}{3}\right) + \frac{1}{2} \sin\left(x - \frac{\pi}{3}\right) \\
 &= \sin \frac{\pi}{3} \cos\left(x - \frac{\pi}{3}\right) + \cos \frac{\pi}{3} \sin\left(x - \frac{\pi}{3}\right) \\
 &= \sin\left(\frac{\pi}{3} + x - \frac{\pi}{3}\right) = \sin x \quad (\text{참})
 \end{aligned}$$

$$\begin{aligned}
 \text{B. } & (\sin^{-1}x)' = \frac{1}{\sqrt{1-x^2}} \text{이므로} \\
 \sin^{-1}x &= \int \frac{1}{\sqrt{1-x^2}} dx = \int_0^x \frac{1}{\sqrt{1-t^2}} dt \\
 &= \int_0^x (1-t^2)^{-\frac{1}{2}} dt \\
 &= \int_0^x \sum_{n=0}^{\infty} \binom{-\frac{1}{2}}{n} (-1)^n t^{2n} dt \quad \text{이다. (참)}
 \end{aligned}$$

$$\begin{aligned}
 \text{C. } & 0 < x < 4 \Leftrightarrow -2 < x-2 < 2 \Leftrightarrow -1 < \frac{x-2}{2} < 1 \text{ 이므로} \\
 \frac{1}{4-x} &= \frac{1}{2-(x-2)} = \frac{1}{2} \times \frac{1}{1-\frac{x-2}{2}} \\
 &= \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{x-2}{2}\right)^n = \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} (x-2)^n \quad \text{이다. (참)}
 \end{aligned}$$

정답 : ④

$$\begin{aligned}
 \text{[2] } & f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots \\
 &= x + x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots \\
 &\quad (\text{조건 (가) } f(0) = 0, f'(0) = 1, f''(0) = 2 \text{를 적용}) \\
 &= x + x^2 + \sum_{n=3}^{\infty} \frac{f^{(n)}(0)}{n!} x^n \\
 &\leq x + x^2 + \sum_{n=3}^{\infty} \frac{n!}{n!(n^2-3n+2)} x^n \quad (\text{조건 (나)를 적용}) \\
 &= x + x^2 + \sum_{n=3}^{\infty} \frac{1}{(n-1)(n-2)} x^n \text{이므로} \\
 f(1) &\leq 1 + 1 + \sum_{n=3}^{\infty} \frac{1}{(n-1)(n-2)} = 2 + \sum_{n=3}^{\infty} \left\{ \frac{1}{n-2} - \frac{1}{n-1} \right\} \\
 &= 2 + \lim_{n \rightarrow \infty} \left\{ \left( \frac{1}{1} - \frac{1}{2} \right) + \left( \frac{1}{2} - \frac{1}{3} \right) + \dots + \left( \frac{1}{n-2} - \frac{1}{n-1} \right) \right\} \\
 &= 2 + \lim_{n \rightarrow \infty} \left( 1 - \frac{1}{n-1} \right) = 3 \quad \text{이다.} \\
 &\text{즉, } f(1) \leq 3 \text{이므로 } m \text{의 최솟값은 } 3 \text{이다.}
 \end{aligned}$$

정답 : ③

$$\begin{aligned}
 \text{[3] } & f(x) = \begin{cases} \frac{1 - \cos x}{x^2}, & x \neq 0 \\ \frac{1}{2}, & x = 0 \end{cases} \\
 &= \begin{cases} \frac{1 - \left(1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots\right)}{x^2}, & x \neq 0 \\ \frac{1}{2}, & x = 0 \end{cases}
 \end{aligned}$$

$$= \frac{1}{2!} - \frac{1}{4!}x^2 + \frac{1}{6!}x^4 - \frac{1}{8!}x^6 + \frac{1}{10!}x^8 - \frac{1}{12!}x^{10} + \dots$$

$$\text{이므로 } f^{(8)}(0) + f^{(9)}(0) + f^{(10)}(0) = \frac{8!}{10!} - \frac{10!}{12!} \text{이다.}$$

정답 : ③

$$\text{[4] } f(x) = \sqrt{1+x} \text{일 때,}$$

$$f'(x) = \frac{1}{2\sqrt{1+x}} = \frac{1}{2}(1+x)^{-\frac{1}{2}}, \quad f''(x) = -\frac{1}{4}(1+x)^{-\frac{3}{2}} \text{이다.}$$

따라서

$$f'(x) = f(0) + f'(0)x + \frac{1}{2!}f''(0)x^2 + \dots \text{이므로}$$

$$T_1(x) = f(0) + f'(0)x \text{라 하면,}$$

$$R_1(x) = \frac{1}{2!}f''(0)x^2 + \dots \text{이므로 테일러 부등식에 의하여}$$

$$\begin{aligned}
 |R_1(x)| &= \left| \frac{1}{2!}f''(c)(c-0)^2 \right| \\
 &= \left| \frac{1}{2!} \times -\frac{1}{4} \times \frac{1}{(1+c)^{\frac{3}{2}}} (c-0)^2 \right| \\
 &= \frac{1}{2} \times \frac{1}{4} \times \left| \frac{1}{(1+c)^{\frac{3}{2}}} (c-0)^2 \right| \\
 &\leq \frac{1}{8} \times \frac{1}{(0.99)^{\frac{3}{2}}} \times (0.01)^2 \quad (\because |c| < 0.01) \text{이다.}
 \end{aligned}$$

정답 : ⑤

$$\begin{aligned}
 \text{[5] } & F(x) = \int_0^x \cos(t^2) dt \\
 &= \int_0^x \left( 1 - \frac{1}{2!}t^4 + \frac{1}{4!}t^8 - \frac{1}{6!}t^{12} + \dots \right) dt \\
 &= x - \frac{1}{2! \times 5}x^5 + \frac{1}{4! \times 9}x^9 - \frac{1}{6! \times 13}x^{13} + \dots
 \end{aligned}$$

$$\text{이므로 } P_5(x) = x - \frac{1}{2! \times 5}x^5 \text{이고,}$$

$$E(x) = F(x) - P_5(x) = \frac{1}{4! \times 9}x^9 - \frac{1}{6! \times 13}x^{13} + \dots \text{이다.}$$

$$\begin{aligned}
 \text{가. } E(0.1) &= \frac{1}{4! \times 9}(0.1)^9 - \frac{1}{6! \times 13}(0.1)^{13} + \dots \text{이므로} \\
 E(0.1) &> 0 \text{이다.}
 \end{aligned}$$

$$\begin{aligned}
 \text{나. } E(x) &= \frac{1}{4! \times 9}x^9 - \frac{1}{6! \times 13}x^{13} + \dots \text{이므로 기함수이므로} \\
 E(-0.1) &= -E(0.1) \text{이고 (가)에 의하여} \\
 E(-0.1) &< 0 \text{이다.}
 \end{aligned}$$

$$\text{다. } E(1) = \frac{1}{4! \times 9} - \frac{1}{6! \times 13} + \dots < \frac{1}{4! \times 9} = \frac{1}{216} < \frac{1}{200}$$

$$\text{라. } P_5(x) = P_6(x) = P_7(x) = P_8(x) \text{이므로}$$

$$\begin{aligned}
 |E(x)| &= |R_8(x)| = \left| \frac{1}{9!}f^{(9)}(c)(c-0)^9 \right| \\
 &\leq \frac{M}{9!}|c-0|^9 \\
 &\quad (\because [-1,1] \text{에서 } |f^{(9)}(x)| \text{의 최댓값은 } M \text{이다.}) \\
 &\leq \frac{M}{9!} \times 1^9 \text{이다.}
 \end{aligned}$$

$$\text{즉, } E(1) \leq \frac{M}{9!}$$

정답 : ⑤

【6】  $na_{n+1} + (\ln 2)a_n = 0$  일 때,

$n = 1$  이면  $a_2 + (\ln 2)a_1 = 0$  이므로  $a_2 = -\ln 2$ ,

$n = 2$  이면  $2a_3 + (\ln 2)a_2 = 0$  이므로  $a_3 = \frac{(-\ln 2)^2}{2!}$ ,

$n = 3$  이면  $3a_4 + (\ln 2)a_3 = 0$  이므로  $a_4 = \frac{(-\ln 2)^3}{3!}$ ,

$\vdots$

이다.

$e^x = 1 + x + \frac{1}{2!}x^2 + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$  이므로

$$\sum_{n=1}^{\infty} a_n = \sum_{n=0}^{\infty} \frac{(-\ln 2)^n}{n!} = e^{-\ln 2} = \frac{1}{2} \text{ 이다.}$$

[다른 풀이]

$na_{n+1} + (\ln 2)a_n = 0$  이므로

$$a_{n+1} = \frac{(-\ln 2)}{n} a_n,$$

$$a_n = \frac{(-\ln 2)}{n-1} a_{n-1},$$

$$a_{n-1} = \frac{(-\ln 2)}{n-2} a_{n-2},$$

$\vdots$

이다.

$$a_{n+1} = \frac{(-\ln 2)}{n} a_n = \frac{(-\ln 2)^2}{n(n-1)} a_{n-1}$$

$$= \frac{(-\ln 2)^3}{n(n-1)(n-2)} a_{n-2}$$

$\vdots$

$$= \frac{(-\ln 2)^n}{n(n-1)(n-2) \times \dots \times 1} a_1$$

$$= \frac{(-\ln 2)^n}{n!} \quad (\because a_1 = 1)$$

$$\text{이므로 } \sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{(-\ln 2)^{n-1}}{(n-1)!} = \sum_{n=0}^{\infty} \frac{(-\ln 2)^n}{n!} \text{ 이다.}$$

$e^x = 1 + x + \frac{1}{2!}x^2 + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$  이므로

$$\sum_{n=1}^{\infty} a_n = \sum_{n=0}^{\infty} \frac{(-\ln 2)^n}{n!} = e^{-\ln 2} = \frac{1}{2} \text{ 이다.}$$

정답 : ②

【7】  $\tan^{-1} x = x - \frac{1}{3}x^3 + \dots = \sum_{n=0}^{\infty} \frac{(-1)^{n-1}}{2n-1} x^{2n-1}$  이므로

양변을 적분하면,

$$\int \tan^{-1} x \, dx = \sum_{n=0}^{\infty} \frac{(-1)^{n-1}}{2n(2n-1)} x^{2n} + C$$

$$\Rightarrow x \tan^{-1} x - \int \frac{x}{1+x^2} dx = \sum_{n=0}^{\infty} \frac{(-1)^{n-1}}{2n(2n-1)} x^{2n} + C$$

$$\Rightarrow x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) = \sum_{n=0}^{\infty} \frac{(-1)^{n-1}}{2n(2n-1)} x^{2n} + C$$

이므로  $x = 0$  을 대입하면  $C = 0$  이다.

$$\sum_{n=0}^{\infty} \frac{(-1)^{n-1}}{2n(2n-1)} x^{2n} = x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) \text{ 이므로}$$

$x = \frac{1}{\sqrt{3}}$  을 대입하면,

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n(2n-1)3^n} = \frac{\pi}{6\sqrt{3}} - \frac{1}{2} \ln \frac{4}{3} \text{ 이다.}$$

정답 : ①

$$\text{【8】 } \sum_{n=1}^{\infty} \frac{n^2+n-1}{(n+1)!} = \sum_{n=1}^{\infty} \frac{n(n+1)-1}{(n+1)!}$$

$$= \sum_{n=1}^{\infty} \frac{n(n+1)}{(n+1)!} - \frac{1}{(n+1)!}$$

$$= \sum_{n=1}^{\infty} \frac{1}{(n-1)!} - \frac{1}{(n+1)!}$$

$$= \lim_{k \rightarrow \infty} \sum_{n=1}^k \frac{1}{(n-1)!} - \frac{1}{(n+1)!}$$

$$= \lim_{k \rightarrow \infty} \left( \frac{1}{0!} - \frac{1}{2!} \right) + \left( \frac{1}{1!} - \frac{1}{3!} \right) + \left( \frac{1}{2!} - \frac{1}{4!} \right) + \left( \frac{1}{3!} - \frac{1}{5!} \right)$$

$$+ \dots + \left( \frac{1}{(k-2)!} - \frac{1}{k!} \right) + \left( \frac{1}{(k-1)!} - \frac{1}{(k+1)!} \right)$$

$$= \lim_{k \rightarrow \infty} \frac{1}{0!} + \frac{1}{1!} - \frac{1}{k!} - \frac{1}{(k+1)!}$$

$$= 2$$

[다른 풀이]

$$e^x - 1 - x = \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \dots$$

$$= \sum_{n=2}^{\infty} \frac{1}{n!} x^n = \sum_{n=1}^{\infty} \frac{1}{(n+1)!} x^{n+1}$$

을 양변 미분하면  $e^x - 1 = \sum_{n=1}^{\infty} \frac{(n+1)}{(n+1)!} x^n$  이고,

다시 양변을 미분하면  $e^x = \sum_{n=1}^{\infty} \frac{(n+1)n}{(n+1)!} x^{n-1}$  이다.

$x = 1$  을 대입하면  $e = \sum_{n=1}^{\infty} \frac{(n+1)n}{(n+1)!}$  이다.

또한  $e^x - 1 - x = \sum_{n=1}^{\infty} \frac{1}{(n+1)!} x^{n+1}$  이므로

$x = 1$  을 대입하면  $e - 2 = \sum_{n=1}^{\infty} \frac{1}{(n+1)!}$  이다.

$$\text{즉, } \sum_{n=1}^{\infty} \frac{(n+1)n-1}{(n+1)!} = e - (e-2) = 2$$

정답 : ②

【9】  $\sum_{n=1}^{\infty} n^2 x^n = \frac{x+x^2}{(1-x)^3}$  을 양변 미분하면

$$\sum_{n=1}^{\infty} n^3 x^{n-1} = \frac{(1+2x)(1-x)^3 + 3(x+x^2)(1-x)^2}{(1-x)^6}$$

$$= \frac{(1+2x)(1-x) + 3(x+x^2)}{(1-x)^4}$$

$$= \frac{1+4x+x^2}{(1-x)^4}$$

이므로  $\sum_{n=1}^{\infty} n^3 x^n = \frac{x+4x^2+x^3}{(1-x)^4}$  이다.

$$\text{즉, } \sum_{n=1}^{\infty} n^3 \left( \frac{1}{2} \right)^n = \frac{\frac{1}{2} + 4 \left( \frac{1}{2} \right)^2 + \left( \frac{1}{2} \right)^3}{\left( 1 - \frac{1}{2} \right)^4} = 26$$

정답 : ④

【10】  $\sum_{n=1}^{\infty} \frac{x^n}{n(n+1)} = \frac{(1-x)\ln(1-x)+x}{x}$  이고,

$\sum_{n=2}^{\infty} \frac{x^n}{n(n+1)} = \frac{(1-x)\ln(1-x)+x}{x} - \frac{x}{2}$  이다.

따라서

$$\begin{aligned} & \sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} \left(\frac{1}{2}\right)^{n-1} \\ &= \sum_{n=1}^{\infty} \frac{2}{n(n+1)(n+2)} \left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} \frac{(n+2)-n}{n(n+1)(n+2)} \left(\frac{1}{2}\right)^n \\ &= \sum_{n=1}^{\infty} \left[ \frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right] \left(\frac{1}{2}\right)^n \\ &= \sum_{n=1}^{\infty} \frac{1}{n(n+1)} \left(\frac{1}{2}\right)^n - \sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)} \left(\frac{1}{2}\right)^n \\ &= \sum_{n=1}^{\infty} \frac{1}{n(n+1)} \left(\frac{1}{2}\right)^n - \sum_{n=2}^{\infty} \frac{1}{n(n+1)} \left(\frac{1}{2}\right)^{n-1} \\ &= \sum_{n=1}^{\infty} \frac{1}{n(n+1)} \left(\frac{1}{2}\right)^n - 2 \sum_{n=2}^{\infty} \frac{1}{n(n+1)} \left(\frac{1}{2}\right)^n \\ &= \frac{\frac{1}{2} \ln \frac{1}{2} + \frac{1}{2}}{\frac{1}{2}} - 2 \left[ \frac{\frac{1}{2} \ln \frac{1}{2} + \frac{1}{2}}{\frac{1}{2}} - \frac{\frac{1}{2}}{2} \right] \\ &= \ln \frac{1}{2} + 1 - 2 \left( \ln \frac{1}{2} + \frac{3}{4} \right) = \ln 2 - \frac{1}{2} \text{ 이다.} \end{aligned}$$

정답 : ①



장황수학